Using the results of [257, Lemma 3.4], we can deduce that for any $\delta > 0$, there exists a sufficiently small $\varepsilon > 0$ such that if a realization of the Wiener process $\mathbf{W}$ remains close to our solution $\boldsymbol{R}$

$$\sup_{0 \le s \le t} \|\mathbf{W}(s) - \boldsymbol{R}(s)\| \le \varepsilon, \tag{6.50}$$

then $\mathbf{X}(t) \in \mathcal{B}_\delta(\boldsymbol{y})$. That is, if the realized Wiener process stays within a tube of radius ε centered on our derived solution $\boldsymbol{R}$ (for some sufficiently small ε), then our solution at time t is at most δ from the point $\boldsymbol{y}$.

The event (6.50) is shown in [349] to have positive probability for any $\varepsilon > 0$, and hence the required property is shown for any open set.

As Lemma 6.1 holds for all open subsets C, Assumption 1(i) holds as a corollary. Thus we have demonstrated that all of the necessary assumptions hold for Theorem 6.2. Hence Langevin dynamics is geometrically ergodic with respect to the canonical distribution

$$\rho_\beta(\boldsymbol{q},\boldsymbol{p}) = \exp(-\beta H(\boldsymbol{q},\boldsymbol{p})),$$

where it is easily checked that

$$\mathcal{L}_{\mathrm{LD}}^\dagger \rho_\beta = 0.$$

Hence averages along solutions to (6.47) will converge to the spatial average with respect to ρ_β. Thus, Langevin dynamics is a useful tool for sampling the canonical distribution, as we may sample ρ_β by computing solutions to the SDE. This remains non-trivial, as the problem usually exists in a very high dimensional space (we may consider millions of particles) with complicated non-linear potential energy function $U(\boldsymbol{q})$. Many algorithms have been proposed for computing solutions to Langevin dynamics [46, 55, 62, 265, 311]. We shall compare the sampling accuracy of some of these schemes, and address strategies for the development of discretization methods for Langevin dynamics in the following chapter.

We may also investigate the properties of the operator $\mathcal{L}_{\mathrm{LD}}$, just as we did for Brownian dynamics. However for Langevin dynamics, the lack of coupling between the noise process and all phase variables makes it relatively difficult to prove ergodicity. We denote by $\mathcal{H}^1(\mu)$ the space of functions $f(\boldsymbol{q},\boldsymbol{p})$ defined on $\mathcal{D} = \mathcal{T}^{\mathrm{N}_\mathrm{c}} \times \mathbb{R}^{\mathrm{N}_\mathrm{c}}$ with bounded norm

$$\|f\|^2_{\mathcal{H}^1(\mu)} \stackrel{\mathrm{def}}{=} \|f\|^2_{L^2(\mu)} + \|\nabla_{\boldsymbol{q}} f\|^2_{L^2(\mu)} + \|\nabla_{\boldsymbol{p}} f\|^2_{L^2(\mu)},$$

where

$$\|f\|^2_{L^2(\mu)} = \int_{\mathcal{D}} f^2 \mu(\mathrm{d}\omega),$$

and, for a vector function $\boldsymbol{w} : \mathcal{D} \to \mathbb{R}^{\mathrm{N_c}}$, we have

$$\|\boldsymbol{w}\|^2_{L^2(\mu)} = \sum_{i=1}^{\mathrm{N_c}} \|w_i\|^2_{L^2(\mu)}.$$

With this in mind, we state a useful property that we will need in the next chapter in order to clarify results on the error in invariant measures.

Proposition 6.4 *Let U be C^∞ on the position domain $\mathcal{Q} = \mathcal{T}^{\mathrm{N_c}}$ and suppose $g \in \mathcal{H}^1(\mu)$ satisfies*

$$\int_{\mathcal{D}} g \mathrm{d}\mu_\beta = 0,$$

where μ_β is the canonical measure, then the Poisson equation

$$\mathcal{L}^\dagger_{\mathrm{LD}} \varphi = g$$

has a solution $\varphi \in \mathcal{H}^1(\mu)$ which is unique up to addition of a constant multiple of the Gibbs density.

The condition is a consequence of the Fredholm Alternative which says that, for a linear operator $\mathcal{A}$ with compact resolvent, we have one of the following two conditions:

- $\mathcal{A}\varphi = g$ has a solution.
- There is a function y with $\mathcal{A}^\dagger y = 0$ and $\langle y, g \rangle \neq 0$.

In our case, $\mathcal{A} = \mathcal{L}^\dagger_{\mathrm{LD}}$. The only solutions of $\mathcal{L}_{\mathrm{LD}} y = 0$ (conserved quantities of Langevin dynamics) are the constants. So the condition of the proposition eliminates the second alternative, leaving only the first.

If we have two solutions φ_1, φ_2 of $\mathcal{L}^\dagger_{\mathrm{LD}} \varphi = g$, then their difference w must be a solution of $\mathcal{L}^\dagger_{\mathrm{LD}} w = 0$ and the only such solution is a constant multiple of the Gibbs density ρ_β.

More properties of the operator $\mathcal{L}_{\mathrm{LD}}$ are collected in [223].

6.4.5 Convergence of Averages

Given a collection of phase space points distributed according to the density $\rho_0(\boldsymbol{q}, \boldsymbol{p})$, we know that its evolution (under Langevin dynamics) after a time t will be described by

$$\rho(\boldsymbol{q}, \boldsymbol{p}, t) = e^{t\mathcal{L}^\dagger_{\mathrm{LD}}} \rho_0(\boldsymbol{q}, \boldsymbol{p}).$$